

Example exercise 3 bis

Wednesday, 29 April 2020

11.29

Determine whether the following two CCS expressions

$$a.(b.Nil + c.Nil) \quad \text{and} \quad a.(b.Nil + \tau.(b.Nil + c.Nil))$$

are :

- strongly bisimilar?
- weakly bisimilar?

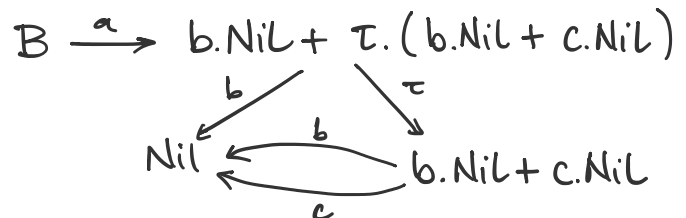
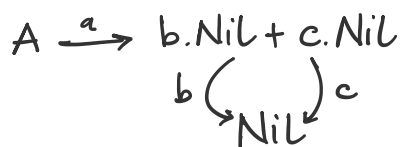
Solution: The processes above are not strongly bisimilar but they are weakly bisimilar.

Let

$$A = a.(b.Nil + c.Nil) \quad \text{and}$$

$$B = a.(b.Nil + \tau.(b.Nil + c.Nil))$$

The LTS generated by A and B are as follows:



- We can prove that $A \not\sim B$ by providing a distinguishing formula :

$$A \not\models \langle a \rangle \langle \tau \rangle tt$$

$$B \models \langle a \rangle \langle \tau \rangle tt$$

Then, by Hennessy-Milner theorem, we know that $A \not\sim B$.

- We can prove $A \approx B$ by providing a weak bisimulation relation R containing the pair (A, B) .

$$R = \{ (A, B), (b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil})), (b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + c.\text{Nil}), (\text{Nil}, \text{Nil}) \}$$

The relation above is a bisimulation relation because the following conditions hold:

- * if $A \xrightarrow{a} b.\text{Nil} + c.\text{Nil}$, then $B \xRightarrow{a} b.\text{Nil} + c.\text{Nil}$ and $(b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + c.\text{Nil}) \in R$
- * if $(b.\text{Nil} + c.\text{Nil}) \xrightarrow{b,c} \text{Nil}$, then $(b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil})) \xRightarrow{b,c} \text{Nil}$ and $(\text{Nil}, \text{Nil}) \in R$
- * the last two pairs are clear

And, vice versa

- * if $B \xRightarrow{a} b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil})$ then $A \xRightarrow{a} b.\text{Nil} + c.\text{Nil}$, and $(b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil})) \in R$
- * if $b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil}) \xrightarrow{\tau} b.\text{Nil} + c.\text{Nil}$ then $b.\text{Nil} + c.\text{Nil} \xrightarrow{\tau} b.\text{Nil} + c.\text{Nil}$ and $(b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + c.\text{Nil}) \in R$.
- * if $b.\text{Nil} + \tau.(b.\text{Nil} + c.\text{Nil}) \xrightarrow{b} \text{Nil}$ then $b.\text{Nil} + c.\text{Nil} \xrightarrow{b} \text{Nil}$ and $(\text{Nil}, \text{Nil}) \in R$
- * the last two pairs are clear.

Remark: notice that we could have proven $A \approx B$ by providing a universal winning strategy for defender in a weak bisimulation game.